**James Rocco Quantitative Data Research Scholarship Proposal**

# Analyzing the Characteristics of Institutional Closures Since 2016

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## Abstract

The upcoming *enrollment cliff* due to demographic shifts and economic factors is expected to cause a 15% decline in U.S. higher education enrollment in 2025, varying by institution. We'll study closures so far using the Integrated Postsecondary Education Data System (IPEDS) and a dataset of campus closures and mergers by Prof. Jamshidi. Analysis will include partial endowment data from the NACUBO-TIAA Study of Endowments (NTSE). Statistical analysis, PCA, and machine learning models like k-means clustering, LDA, kNN, decision trees, random forests, and gradient boosting will be used. Finally, a Bayesian hierarchical model will be constructed based on findings to generate future scenarios. We will use R and Python for this project. We will follow a 10-week timeline with proposed deliverables being (1) a publicly available repository of code, data, and findings, (2) a slide show presentation, and (3) an academic paper submission to the *Socially Responsible Modelling, Computation, and Design*. These findings will likely be informative for education leaders and policymakers aiming to understand and mitigate enrollment cliff risks and promote sustainability.

## Research Question and Relevance

The *enrollment cliff* refers to a significant decline in student enrollment that many educational institutions, particularly universities and colleges, are currently experiencing. According to Barshay (2018), college student populations are projected to decrease by more than 15% after the year 2025, a trend that is already evident to academic institutions. This decline is not expected to be uniform across all institutions; demand for elite colleges is anticipated to increase, while overall demand, when analyzed by state, is expected to drop for all but a few exceptions (see Figure 1). Of course, a complicating factor is that some students travel far from home while others choose to remain close. In addition, while the population is dropping, the number of people interested in degrees has continually risen. Paradoxically, we expect the demand on elite institutions to increase, as noted in Harshay 2018. Still, the anticipated hardest-hit states are those whose higher educational institutions per capita is larger, such as Massachusettes, even though these states are often home to such institutions.

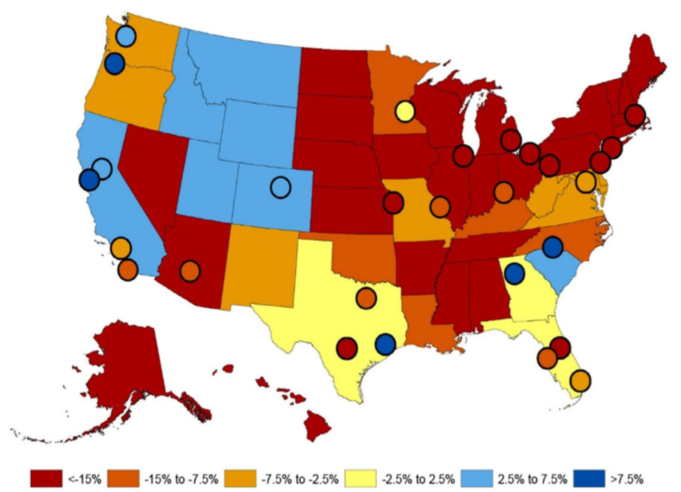

*Figure 1: Forecasted population shifts 2012-2029, from Gawe (2018)*

All in all, administrators are bracing themselves for a turbulent adjustment. Yet it is important to note that the incoming student populations have already been declining and COVID-19 has accelerated this process, as noted by Lo and Young (2021).

As a result, we are interested in the following question: **What can be deduced from the closures that have already happened?** Recent studies have documented an acceleration in college closures, with many institutions ceasing operations between 2016 and 2023 (Gardner, 2023). In particular, we are interested in comparing the population of closed institutions against all others. What defines that? What factors characterize them better than others?

The significance of the findings in this work is evident as it pertains to the overall health of the market of higher education. According to the latest data from the U.S. Department of Commerce's Bureau of Economic Analysis (BEA), education-related travel exports were valued at $50.2 billion and ranked 7th among service exports in 2023. The United States hosted 1,126,690 international students in the 2023-2024 academic year, reflecting a six percent increase over the previous year, as reported by IIE's Open Doors 2024 Report. Furthermore, HolonIQ forecasts that global education and training expenditure will reach $10 trillion by 2030. So this is a crucial market in the US and one growing internationally.

As discussions around trade deficits continue to dominate U.S. international policy and news cycles, it becomes clear that institutions of higher education play a crucial, yet often overlooked, role in addressing these issues. Therefore, the health and stability of this market are vital to the nation's economic well-being. Our work will be one small component in assisting administrators and policy makers to make more informed decisions.

## Data

The Integrated Postsecondary Education Data System (IPEDS) is a database hosted by the Department of Education in the US. This database includes several features of campuses including address, degrees offered, student body size, location classification (e.g., Suburb:Large), financial aid, athletics, accreditation, carnegie classification, tuition, expenses and rates of retention. For this project, this will be our primary source of institutional features as it is comprehensive.

In addition, Prof. Jamshidi has meticulously compiled a dataset of campus closures and mergers from various sources, including Google News alerts, Higher Ed Dive, BestColleges.com, and her professional network. Our analysis focuses on the 2016-2025 period and exclusively on nonprofit institutions, as we consider for-profit institutions to be a distinct market.

A critical gap in both the IPEDS data and our proprietary dataset is the lack of comprehensive endowment information, which is essential for assessing an institution's financial health. We have partial data from the NACUBO-TIAA Study of Endowments (NTSE) dating back to 2019. This dataset provides critical financial metrics for participating institutions in the most recent study year, including detailed endowment performance metrics, spending rates, and investment strategies. Preliminary cross-referencing between IPEDS and NTSE shows a very high match rate for institutional identifiers, requiring minimal reconciliation effort. This allows for preliminary analysis, and we intend to leverage this data to infer the role of endowments in institutional closures.

To address missing data, we may explore imputation methods using publicly available datasets such as DataUSA.io and Wikipedia, both of which include endowment information. However, the reliability of these sources is uncertain. If used, we would validate their figures against known data from NTSE and make necessary adjustments. While this approach will provide insights into the influence of endowments, it may also introduce limitations to our findings.

## Technical Approach

Once constructed, the dataset will consist of several features and will likely require value imputation, particularly for data related to endowments. This will necessitate a thorough statistical analysis of feature correlations. Conditional means and linear regression will be employed to estimate missing values, which will then be cross-referenced against external sources such as DataUSA.io for validation.

Additional statistical analyses will be conducted to identify which features are most influential in predicting the binary outcome (closure versus non-closure), likely involving multiple t-tests. Principal Component Analysis (PCA) will be implemented to assess the informational content of the features and explore potential data compression and its implications. It will be crucial to determine whether the new basis elements computed by PCA correspond to meaningful underlying factors or are merely mathematical artifacts. Additionally, the Variance Inflation Factor (VIF) will be used to identify and address multicollinearity among features, refining the dataset accordingly.

Next, a metric will be designed to characterize the "distance" between samples in the dataset. This phase will involve making value judgments regarding sample similarity and selecting appropriate functions to quantify this. Feature scaling will also be determined during this phase.

Several standard machine learning models will then be implemented. Initially, k-means clustering will be used to identify general groupings of institution types, irrespective of closures. Cross-validation will be employed to select hyperparameters, and the previously defined metric will be utilized. The resulting clusters will be evaluated for meaningfulness by comparing them to established classifications like the Carnegie Classification. Success in this phase will help identify the most vulnerable institution types.

Linear Discriminant Analysis (LDA) will follow, assessing the separability of the dataset's labels. Success here will indicate a clear distinction between institutions that have closed and those that have not, providing valuable insights into these differences.

Subsequently, various models including k-nearest neighbors (kNN), decision tree, random forest, and gradient boosting will be tuned to estimate how changes in specific features affect the probability of closure. The outcomes of the previous models will either support or challenge these findings. If successful, these models will help characterize factors that influence the likelihood of closure.

Finally, a Bayesian hierarchical model will be developed based on key features identified through earlier analyses. This model, designed for multi-level random processes, will consider the most influential factors in institutional closures and generate potential future scenarios.

R and Python will serve as the primary programming languages due to their robust libraries for statistical modeling and machine learning. Key packages will include `arm` in R, and `statsmodels` and `scikit-learn` in Python.

## Expertise

Throughout Akbari's tenure at Lake Forest College, he has completed coursework that has prepared him for the analytical and modeling demands of this research project. Courses such as Computational Math (CSCI 240), Statistics and Statistical Programming with R (MATH 150, 250), and Machine Learning (CSCI 250, 450) have equipped Akbari with the skills necessary to handle and process data using a variety of techniques. These courses have enabled Akbari to perform statistical analyses and tests, apply mathematical imputation methods, implement algorithms for data processing, and utilize machine learning models for prediction and classification tasks.

Beyond coursework, Akbari has engaged in faculty-led research projects. His current project focuses on statistical testing of sequences using R, incorporating feature vector construction for classification tasks. The project analyzes over 10 aspects of randomness, constructs feature vectors, and applies PCA for dimensionality reduction to assess feature impact on predictability. Using a classification model, any sequence of numbers can be labeled by a randomness generation method. Other than a comprehensive open-source repository and summary paper, the project results in a tool for researchers to test the predictability of sequences being used in simulation models. This project will conclude by May 1st 2025, before the start of the James Rocco Program.

This project is being supervised by Dr. Sara Jamshidi. Dr. Jamshidi has been in the sphere of machine learning since 2015, where she has worked on reasoning models for LIDAR detectors, proprioception for submersibles, and triangulation with sound arrays. In addition, she has ongoing work developing new ways of finding solutions to systems of polynomial equations, a computationally intractable problem (often EXSPACE-hard) undergirding much of science that leverages machine learning and strategic biased sampling. As an assistant professor in computer science and mathematics, she teaches Artificial Intelligence, Linear Algebra, Computational Mathematics, Programming for Data Applications, Abstract and Discrete Mathematics, and Computer Science I and II. In addition, she teaches two Coursera courses in Statistical Learning and Computational Bayesian Statistics for the Master's in Data Science program at Illinois Tech. She is also a subject matter expert in Artificial Intelligence and Red Teaming with TrustVector, a start up providing AI Auditing solutions, especially within the sphere of healthcare. She is an active researcher in the intersection of machine learning and computational algebraic geometry, with pending grants in collaboration with major institutions including Argonne National Lab, the University of Notre Dame, the University of Arizona, the University of Hawai'I, and the Illinois Institute of Technology.

She has advised 14 student projects during her time at Lake Forest College, all in areas related to data science and AI.

## Broader Impacts

The proposed research project offers several meaningful impacts across academic, professional, and societal dimensions. At its core, this work enhances our understanding of institutional sustainability in higher education during a period of significant demographic change. The findings will provide actionable insights for educational leaders and policymakers striving to preserve access to higher education, particularly in regions at risk of becoming educational deserts.

From a professional development perspective, this project aligns with Akbari's career aspirations of pursuing a doctorate in Computer Science with a focus on machine learning. Engaging in research with a strong social impact component aligns with Akbari's ideal of using mathematical and computational knowledge to drive positive societal change. This research serves the public good by helping to preserve educational opportunities in vulnerable communities. By identifying early indicators of institutional distress, our work could help prevent sudden college closures that adversely affect students, faculty, and local communities.

Furthermore, the project has strong potential for expansion into a larger research initiative, particularly by incorporating additional data sources or extending the analysis to explore intervention strategies for at-risk institutions. The Bayesian modeling framework developed in this research could be adapted to create early warning systems for institutional stress, serving as a valuable tool for educational administrators and accrediting bodies.

## Deliverables and Timeline

The project will follow a carefully planned 10-week schedule that balances data preparation, analysis, and documentation. Each phase includes specific milestones to ensure steady progress toward our research objectives.

- **Week 1-2:**
  - Data cleaning and organization for closed institutions
  - Imputations
  - Statistical analyses of data
  - Literature review
- **Week 3-6:**
  - Development of models outlined in Tehcnical Approach
  - Drafting introduction and initial findings of paper
- **Week 7-9:**
  - Additional, unanticipated work of improved modelling
  - Completing paper draft.
- **Week 10:**
  - Finalizing the paper for submission;
  - preparing the presentation.

*Final Deliverables:*

The project will produce three primary outputs:

- A **research paper** submitted to *Socially Responsible Modelling, Computation, and Design*.
- A comprehensive **open-source repository** including the constructed dataset, reproducible code and results.
- A **formal presentation** summarizing findings and implications.

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